Haoran Xu

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Education

Zhejiang University 2024 – June 2028 (expected)

Undergraduate, Control Science and Engineering

GPA: **4.22/5.00** | Weighted Average: **87.30/100**

TOEFL iBT: 5.5/6.0 (2026, new scoring scale)

Research Experience

Machine Learning Research 2026 – Present

Formulated research questions and led their investigation through analytical and empirical work, experimental design, interpretation of results, and manuscript preparation.

- Combined mathematical analysis and controlled experiments to investigate learning procedures and model behavior. Designed comparisons and ablation studies to examine the relationship between analytical expectations and empirical observations.
- Used model interventions and behavioral evaluation to assess what experimental results reveal about a learning system. Compared model behavior across settings to examine the scope of an interpretation.
- Designed quantitative evaluations using statistical comparisons, robustness checks, and sensitivity analyses. Examined the consistency of conclusions across experimental conditions and the limitations of evaluation choices.

Public preprint: Haoran Xu. *Same Data, Different Domains: Representation Dependence in Adaptive Data Mixing*. Preprint, 2026. doi:10.5281/zenodo.23428428

Machine Learning for Bioprocess Soft Sensing

Zhejiang University | Supervised by Prof. Xinggao Liu

- Studied machine learning-based soft sensing using industrial multivariate time series from erythromycin fermentation processes. Explored regression models for predicting process variables from production data.
- Compared prediction errors across target variables and examined changes associated with time, production conditions, and data distributions. Analyzed distribution shifts to investigate the reliability and limitations of regression predictions across process settings.

Honors & Awards

Second-Class Comprehensive Scholarship, Zhejiang University.